ADGSTUDIOS

DisputeGuard

Engine

Test Report

Automated verification of the equations and flows

PASS 34 / 34 checks passed

Generated October 2026 — Rust reference engine

Companion to the DisputeGuard Equations and Algorithms guide.

Summary

Metric	Value
Total checks	34
Passed	34
Failed	0
Sections	7

This report is produced by the `report` binary in the `disputeguard-engine` crate. Each row runs a deterministic check against the reference implementation and records the observed value.

Money and Exact Amounts 5/5

Amounts are whole minor units; precision follows the currency and floating-point rounding never appears.

Check	Observed	Result
R1234.56 parses to cents	`minor = 123456`	**PASS**
0.10 + 0.20 is exact	`minor = 30 (= 0.30)`	**PASS**
JPY rejects fractional input	`parse("1.5", JPY) = BadPrecision`	**PASS**
BHD keeps three decimals	`minor = 1234`	**PASS**
Cross-currency add is refused	`ZAR + USD = CurrencyMismatch`	**PASS**

Ownership and Permission 4/4

View depends on ownership or an analyst/admin role; dispute creation always requires ownership.

Check	Observed	Result
Owner can view	`CanView(owner, t) = true`	**PASS**
Stranger cannot view	`CanView(stranger, t) = false`	**PASS**
Analyst can view	`role grants CanView`	**PASS**
Only owner can create	`analyst cannot create a dispute`	**PASS**

Dispute Lifecycle and Concurrency 4/4

Only authorised transitions proceed; stale versions are rejected and successful changes bump the version.

Check	Observed	Result
SUBMITTED + BEGIN_REVIEW -> UNDER_REVIEW	`authorised transition`	**PASS**
SUBMITTED + ISSUE_REFUND -> bottom	`refund is never a lifecycle transition`	**PASS**
Version bumps on success	`v 3 -> 4`	**PASS**

Check	Observed	Result
Stale version is rejected	`expected 2 != actual 3`	**PASS**

Dispute Creation: Idempotency and Uniqueness 4/4

Retries return the original result; at most one active dispute may exist per transaction.

Check	Observed	Result
Retry replays original dispute	`ids 1 == 1, replayed = true`	**PASS**
Key reuse with new body refused	`KeyReuseWithDifferentRequest`	**PASS**
Second active dispute refused	`ActiveDisputeExists`	**PASS**
Active count is 1	`count = 1`	**PASS**

Deterministic Risk Indicators 12/12

Each signal has its own safe guards; a missing input is reported as unavailable rather than treated as low risk.

Check	Observed	Result
Amount signal bounded in [0,1]	`A(9000; x_max=10000) = 0.9886`	**PASS**
Amount signal clamps at scale	`A(1e6; x_max=10000) = 1.0000`	**PASS**
z-score flags R900 outlier	`z = 46.378 (> 3)`	**PASS**
z-score unavailable at zero variance	`constant history -> None`	**PASS**
Robust z (MAD) flags outlier	`z* = 45.023`	**PASS**
MAD = 0 is undefined (no div-by-zero)	`constant history -> None`	**PASS**
Velocity needs positive window	`V(12, 0) = None`	**PASS**
Haversine JHB->CPT ˜ 1262 km	`d = 1261.6 km`	**PASS**
Impossible travel velocity detected	`v_required = 1262 km/h over 1 h`	**PASS**
Combined score in [0,1]	`R = 0.440`	**PASS**
Missing signal refuses a score	`geography = None -> MissingSignal`	**PASS**
Weights must sum to 1	`sum = 1.3 -> InvalidWeights`	**PASS**

Bounded AI Investigation 3/3

The loop gathers only allowlisted, read-only observations and stops on budget or a disallowed tool, leaving the case unchanged.

Check	Observed	Result
Complete run reaches human review	`stage = Review, tool_calls = 3`	**PASS**
Budget stops before 3rd tool call	`stage = Stopped, tool_calls = 2`	**PASS**
Disallowed tool blocked pre-execution	`tool_calls = 1, observations = 1`	**PASS**

Sample Transaction Data 2/2

The demo transactions shared with the web landing page use unique references and exact positive minor units.

Check	Observed	Result
References are unique	`9 transactions, 9 unique ids`	**PASS**
All amounts positive minor units	`every amount_minor > 0`	**PASS**

Appendix: Sample Transactions

The nine demo transactions used as fixtures (amounts shown as exact ZAR from integer cents).

Ref	Merchant	Category	Amount (ZAR)
TX-0042	Atlas Online	Shopping	1249.00
TX-0043	Corner Coffee	Food & drink	68.00
TX-0044	Fresh Market	Groceries	432.50
TX-0051	Studio Supply	Shopping	899.00
TX-0052	Stream House	Subscription	149.00
TX-0053	Cloud Desk	Software	229.00
TX-0061	Coastline Stay	Accommodation	2450.00
TX-0062	City Shuttle	Transport	185.00
TX-0063	Platform Café	Food & drink	96.00

All figures are illustrative. Risk indicators are advisory and never decide a dispute.